

Experimental Study on Geysers in Covered Manholes during Release of Air Pockets in Stormwater Systems

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Abstract: For stormwater systems, geysers in manholes are a risk to pedestrian and vehicle safety and the urban environment. Manhole covers dancing and blowing off are reported worldwide during heavy rainfall and geyser events. These manhole cover motions impact geyser characteristics but are rarely studied. This work experimentally studied geysers in covered manholes induced by air pocket release. Both fixed and mobile covers were tested, and the effects of manhole diameter and cover ventilation size were considered. The geyser regimes are classified into nongeyser, single-geyser, and multigeysers. The mobile covers can dance and blow off, resulting in pressure at the manhole headspace with characteristics of oscillation with high frequency, decline of peak magnitude, and negative pressure occurrence. The maximum positive pressure can be seven times higher for fixed covers than that for mobile covers. Increasing the manhole diameter decreases the number of geysers and the maximum positive pressure for both fixed and mobile covers. Decreasing cover ventilation reduces the number of geysers and increases the maximum positive pressure for the fixed covers, although it has limited effects for mobile covers. DOI: 10.1061/(ASCE)HY.1943-7900.0001978. © 2022 American Society of Civil Engineers.

Author keywords: Air pocket release; Geyser; Manhole cover motion; Stormwater system; Ventilation.

Introduction

Urban stormwater systems, which consist of underground main tunnels, drop shafts, manholes, and pump stations, are used for stormwater conveyance and storage and preventing urban waterlogging. During intense rainfall events, rapid filling of stormwater systems can cause free-surface–pressurized flow with trapped air pockets in main tunnels, potentially resulting in explosive eruptions of water and air–water mixture through manholes to a level above the ground surface, which are referred to as geysers. Geysers are a risk to the safety of pedestrian and vehicular traffic and can cause environmental pollution.

Guo and Song (1991) proposed that pressure oscillations caused by rapid inflows for stormwater systems could result in geysers as oscillations are superimposed on the tunnel water level. More recently, geysers were considered to be attributed to air–water interactions, in particular air pocket entrapment in main tunnels and

release through manholes (Vasconcelos and Wright 2006, 2011; Leon et al. 2019). Field, experimental, and numerical investigations have been conducted focusing on the characteristics of geysers and their relations to structural geometries, inflow scenarios, and air pocket conditions. Wright et al. (2011a, b) studied a series of real geyser events in a stormwater tunnel in Minneapolis; the field data of pressure and velocity within the tunnel were analyzed in combination with video records of the geysers, and the results showed that the piezometric heads adjacent to the manhole never increase to levels approaching the ground surface during the geysers, which needs to be paid special attention for predicting geyser formation. Cong et al. (2017) experimentally observed geysers of air–water mixture caused by the release of air pockets in a simplified stormwater system and concluded that geysers occur when the diameter ratio of manhole to main tunnel is smaller than 0.56. Based on experiments, Huang et al. (2018a) and Wang et al. (2019) classified geyser regimes into gas-flow and surge-type geysers and found that the ejection height of the surge-type geyser is much larger than that of the gas-flow geyser. Chan et al. (2018) and Chagini and Leon (2020) numerically studied geysers induced by air pocket release; the simulated vertical acceleration of air in water columns in manholes confirmed earlier experimental observations by Wright et al. (2011b). Their results suggested that air pocket compression plays a critical role in geyser formation by causing rapid acceleration and expulsion of the water columns.

In practical engineering, manholes always have covers to prevent people, vehicles, and animals from accidentally falling into stormwater systems. Manhole covers and their ventilation holes should affect geyser characteristics; however, they are rarely considered during geyser studies. Huang et al. (2018b) experimentally studied the effects of orifice plates at the manhole top (which can be considered covers) on system pressures during geysers and found that the maximum water head at the horizontal tunnel invert close to the manhole decreases by two-thirds and the pressure oscillation decreases by 95% when the diameter ratio of orifice to manhole is below 25%. Using computational fluid dynamics, Wang and Vasconcelos (2020) evaluated the possibility of manhole cover

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Note. This manuscript was submitted on July 28, 2021; approved on January 18, 2022; published online on February 28, 2022. Discussion period open until July 28, 2022; separate discussions must be submitted for individual papers. This technical note is part of the *Journal of Hydraulic Engineering*, © ASCE, ISSN 0733-9429.

displacement during rapid filling of stormwater systems by investigating pressures underneath covers. They demonstrated that the pressures associated with air pocket release are much larger than those associated with inertial oscillation, and the maximum pressures increase with decreasing manhole diameter and cover ventilation with an increasing initial water level of the manhole. Their achievements need further experimental verification and do not consider the additional ventilation for manholes created by cover motions. Qian and Zhu (2020) established a one-dimensional model to simulate water column movement in manholes with orifice plates driven by air pocket release and indicated that the peak pressures on orifice plates are mainly influenced by initial water column length and air pocket pressure but are minorly affected by the manhole diameter and height. Their model does not consider the formation of an air–water mixture in manholes that is usually observed during geyser experiments.

News stations reported that during stormwater system overburden after heavy rainfall, covers might alternately move up and down at the manhole top (i.e., dancing) or blow off entirely (Holmes 2017; Fox News 2018). The detailed cover motions during geysers remain unclear. Moreover, cover motions change manhole ventilation, which significantly enhances the complexity of geyser behaviors but is rarely considered and studied. Tijsseling et al. (2019) studied the physics of mobile cover dancing induced by the release of air pockets by means of simple spring-mass-dashpot modeling. Their work does not consider cover ventilation and situations of cover blowoff.

This work experimentally studied geysers in covered manholes resulting from the release of air pockets. Fixed and mobile manhole covers were tested and compared, and the effects of manhole diameter and cover ventilation were considered. Geyser behaviors and cover motions were visualized, and pressure evolutions at the manhole headspace were measured and analyzed.

Methodology

The experiments were conducted at the Hydraulic Laboratory of Hohai University, Nanjing, China. Fig. 1 depicts the schematic diagram of an idealized stormwater system model consisting of a constant head tank, horizontal main tunnel, and vertical manhole. Similar stormwater system models were adopted by Cong et al. (2017) and Wang and Vasconcelos (2020), and their results showed

that the release of air pockets in idealized models can reproduce the essential features of geyser behaviors and pressures during actual geyser events. The constant head tank had dimensions of $2.00 \times 1.60 \times 1.20$ m. A pump and labyrinth overflow weir were set inside the tank to provide a constant head of $H_0 = 0.80$ m for the system. The horizontal tunnel was 17.00 m long and $D = 0.150$ m in diameter, and it was connected to the tank at the upstream end and had a closed downstream end. A valve was arranged in the downstream side of the horizontal tunnel to generate an air pocket with a volume of $V_a = 0.032$ m³. A manhole of 2.00 m height (from its top to the tunnel bottom) and variable diameter ($D_r = 0.075$ and 0.150 m) was connected to the tunnel with a T-junction at 8.30 m from the tank.

In actual settings, manhole covers are just placed on the top of manholes without using fixing structures or, in more complicated settings, sit inside a recessed sleeve that doesn't allow the cover to be displaced before manhole headspace air pressurization overcomes the cover weight and sleeve strength. Simulating the sleeve geometry and strength in scaled experiments is difficult. This study tested the conditions of both fixed and mobile covers, which can represent the situations before and after sleeve failure. Fixed covers made of plexiglass were bolted to the top of the manhole. Assuming a hypothetical stormwater system consisting of a horizontal tunnel 1 m in diameter and 15 m underneath the ground surface and manholes of 0.5 to 1 m in diameter (Wright et al. 2011b; Liu et al. 2020; Wang and Vasconcelos 2020), our setup can be considered a 1:7 scaled model. In general, the actual manhole covers are made of cast iron and have a thickness of around 35 mm. Following the geometric similarity, the mobile covers made of cast iron (density of 7,930 kg/m³) were used, and they were 5 mm thick and 1 cm larger than the manholes in diameter so that they could be placed steadily on the top of the manholes. Circular holes with various diameters were set at the cover centers to control cover ventilation. Six dimensionless ventilation areas $A_v/A_s = 0, 0.01, 0.02, 0.05, 0.10$, and 1 were tested, the first and last of which were the extremes of zero ventilation and full ventilation. Herein, A_v and A_s are the areas of ventilation hole and manhole cross-section, respectively. Choosing the range of A_v/A_s was based on analyzing pictures of manhole covers worldwide searched by Google Images as well as the values of A_v/A_s used by Huang et al. (2018b), Wang and Vasconcelos (2020), and Qian and Zhu (2020). Table 1 summarizes the main experimental parameters.

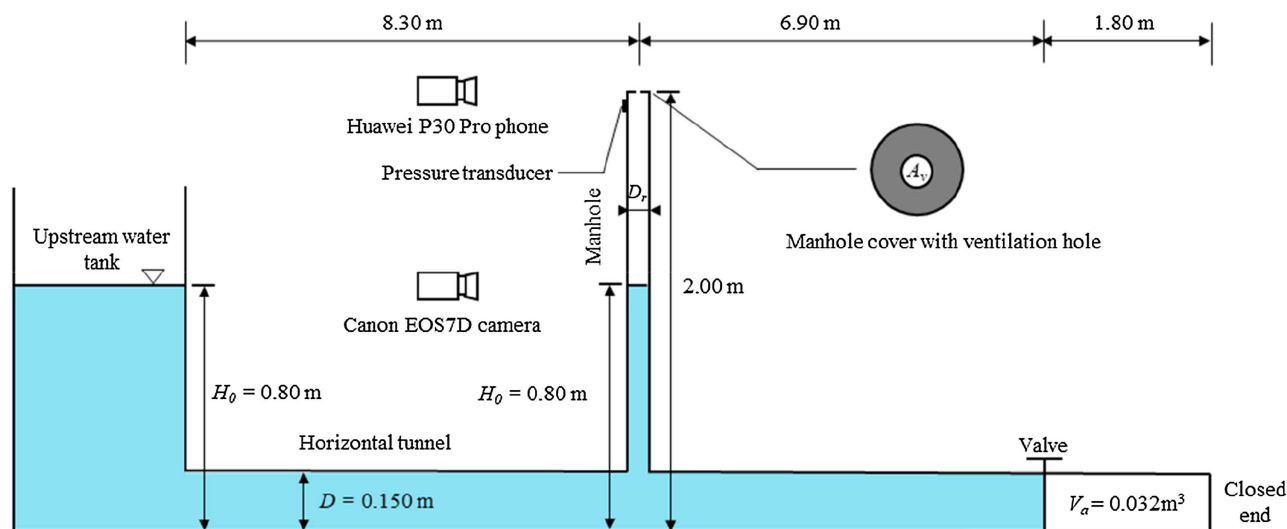


Fig. 1. Schematic diagram of the simplified stormwater system (not to scale).

Table 1. Tested cases

Cases	D_r/D	A_v/A_s	H_0 (m)	V_a (m ³)
Fixed cover	0.5, 1.0	0, 0.01, 0.02, 0.05, 0.1, 1	0.80	0.032
Mobile cover	0.5, 1.0	0, 0.01, 0.02, 0.05, 0.1	0.80	0.032

In a typical experiment, the pump ran continuously, and the model was filled with water to the predetermined level in the tank and the manhole. Then, the valve was closed and the tunnel between the valve and the closed downstream end was emptied, leaving an air pocket at atmospheric pressure. The experiment was started with the valve manually opened quickly, and the average opening valve time was around 0.25 s. The subsequent air pocket movement and potential manhole cover motions were recorded using a Canon EOS7D (Tokyo) camera and a Huawei P30 Pro (Shenzhen, China) phone with 50 and 60 frames per second, respectively. A dynamic pressure transducer (CYG1406F, Shuang-qiao Sensor, Kunshan, China) was mounted at the sidewall of the manhole 2 cm below the cover, and the measurement frequency and range were 128 Hz and from -2 to $+2$ m H₂O, respectively. At least three repetitions were made for each case, and the geyser behavior, manhole cover motions, and features of pressure evolution were similar between the repetitions.

Results and Discussion

Geyser Regimes and Manhole Cover Motions

When the valve is opened, the air pocket propagates in the horizontal tunnel with a smooth nose and slightly turbulent tail. After the air pocket rises into the manhole, three geyser regimes are observed for the fixed cover cases: nongeyser, single-geyser, and multi-geyser. The geyser regimes should depend on system geometries, initial conditions, and inflow scenarios. In our study, the nongeyser regime occurred for the small manhole of $D_r/D = 0.5$ with minor cover ventilation ($A_v/A_s = 0$ and 0.01) as well as the large manhole of $D_r/D = 1$ with all cover ventilation conditions; the single-geyser regime occurred for the small manhole of $D_r/D = 0.5$ with medium cover ventilation ($A_v/A_s = 0.02$, and 0.05), and the multi-geyser regime occurred for the small manhole of $D_r/D = 0.5$ with large cover ventilation ($A_v/A_s = 0.1$, and 1).

For the nongeyser regime, the water column and air pocket behaviors in the manhole are shown in Fig. 2, where t represents the time after opening the valve. The front of the rising air pocket resembled the Taylor bubble, and it slightly pushed the water column upward. The rising distance of the free surface of water column was limited as a large portion of the water volume became the descending film flow near the wall of manhole. The film flow fell, broke, and impacted the flow in the horizontal tunnel, generating an air–water mixture with a frothy whitish color. The air pocket broke in the manhole when the air pocket–water column interface caught up with the free surface of water column. Following the air pocket break, the air–water mixture in the horizontal tunnel entered the manhole and oscillated with damping amplitudes. No water or air–water mixture jetted out of the manhole.

For the multi-geyser regime, the water column and air pocket behaviors in the manhole are shown in Fig. 3. After the air pocket reached the manhole, a part of it rose into the manhole with significant vertical acceleration, pushing the above water column to rapidly eject out of the manhole top (i.e., the first geyser). During the first geyser, the air–water mixture accumulated in the T-junction, and the air pocket in the horizontal tunnel broke into smaller ones. The first geyser caused a sudden pressure drop in the manhole and large pressure gradient between the manhole and tank; therefore, the air pockets and air–water mixture were pushed into the manhole and ascended rapidly, generating an eruption of air–water mixture (i.e., the second geyser). Compared with the first geyser, the second geyser was more violent, with two to three times larger ejection height, and the larger pressure gradient and lower density of the air–water mixture were supposed to be the reasons. After the second geyser, the residual air pockets in the horizontal tunnel continuously rose into the manhole and broke on the free surface, but without generating further geysers.

Leon et al. (2019) also observed multi-geysers in their experiment of air pocket release and stated that the second and third geysers are attributed to water slugs divided by full air segments. These water slugs were not observed in our study. The potential reasons are that the size and pressure of the released air pockets in our study were much smaller than those in Leon et al. (2019), and our study had a constant head tank to continuously supply water into the system during each experimental run. The smaller mass of air pockets and continuous external water supply could prevent the generation of water slugs in the horizontal tunnel.

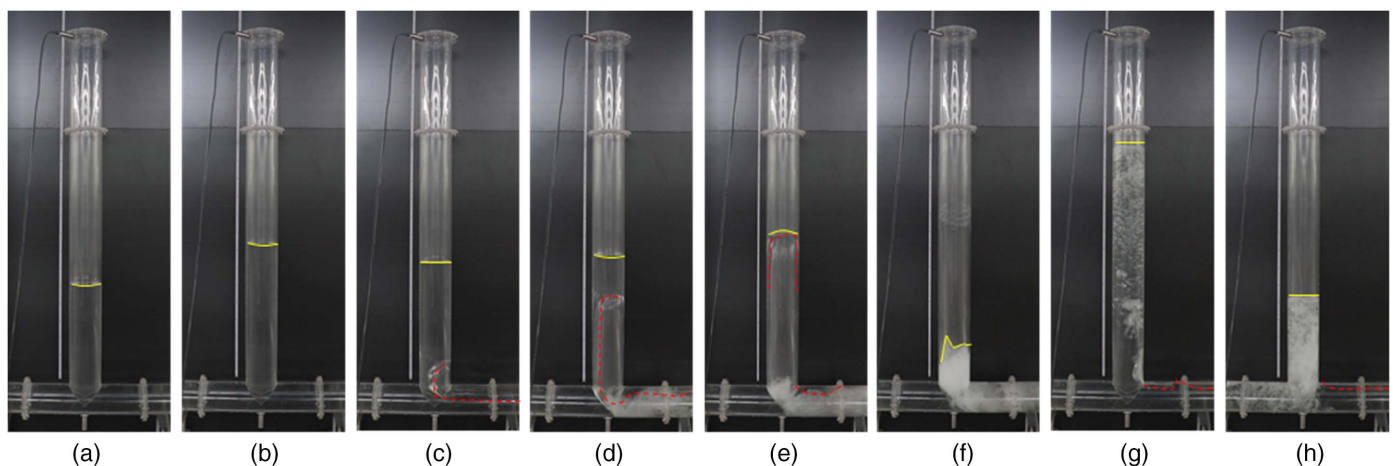


Fig. 2. Water column and air pocket behaviors in manholes with fixed covers during nongeyser regime ($D_r/D = 1$ and $A_v/A_s = 0.05$): (a) $t = 6.22$ s; (b) $t = 8.74$ s; (c) $t = 11.78$ s; (d) $t = 12.40$ s; (e) $t = 12.74$ s (close to air pocket break); (f) $t = 12.98$ s; (g) $t = 15.58$ s; and (h) $t = 18.06$ s; the solid and dashed lines represent the free surface of the water column and the air pocket–water interface respectively.

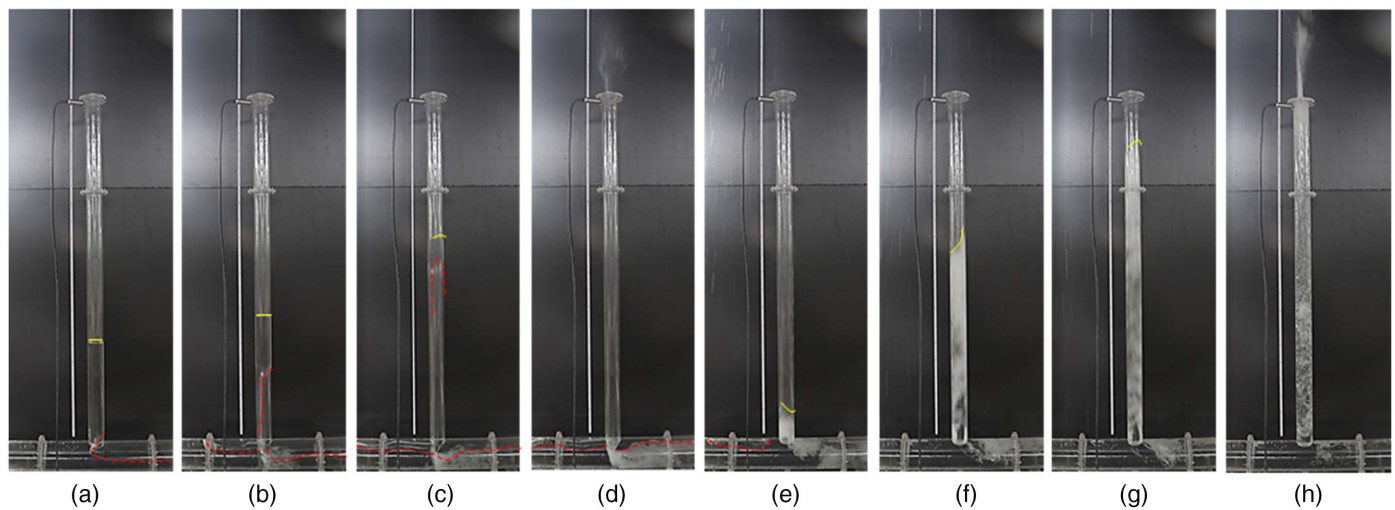


Fig. 3. Water column and air pocket behaviors in manholes with fixed covers during multigeysier regime ($D_r/D = 0.5$ and $A_v/A_s = 0.1$): (a) $t = 11.23$ s; (b) $t = 11.98$ s; (c) $t = 12.46$ s; (d) $t = 13.12$ s (first geyser); (e) $t = 13.68$ s; (f) $t = 14.16$ s; (g) $t = 14.55$ s; and (h) $t = 14.97$ s (second geyser); the solid and dashed lines represent the free surface of the water column and the air pocket–water interface respectively.

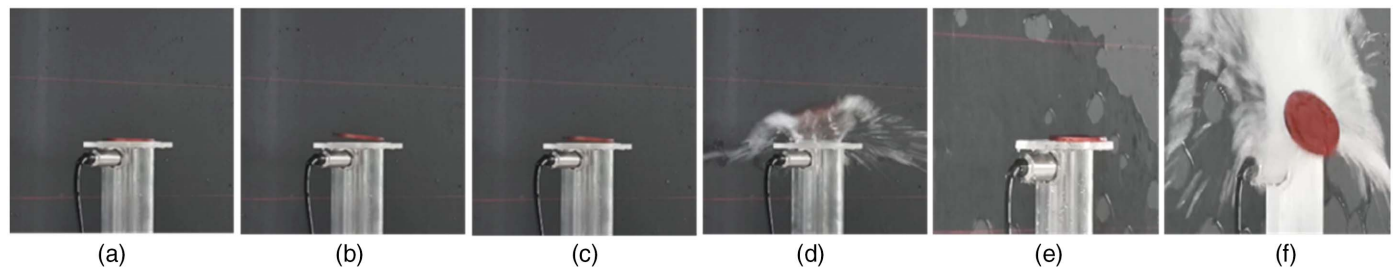


Fig. 4. Photo sequence of mobile cover motions during multigeysier regime ($D_r/D = 0.5$ and $A_v/A_s = 0$): (a) $t = 12.33$ s; (b) $t = 12.71$ s; (c) $t = 13.20$ s; (d) $t = 13.37$ s; (e) $t = 13.95$ s; and (f) $t = 14.62$ s.

The single-geyser regime was very similar to the first geyser of water column ejection caused by the rising air pocket in the multigeysier regime. After the geyser eruption, the air–water mixture entered the manhole and oscillated with attenuation until all the air bubbles escaped from the free surface and the residual water column finally regained stability.

Comparison of the fixed cover cases shows that the manhole cover can decrease the geyser number and intensity, and such effects increase with a decrease in the dimensionless ventilation area A_v/A_s .

For the mobile cover cases, only the nongeyser and multigeysier regimes were observed, which occurred in the large ($D_r/D = 1$) and small manhole ($D_r/D = 0.5$), respectively. The mobile cover ventilation A_v/A_s had a limited effect on geyser regimes.

During the nongeyser regime, slight cover dancing was observed, and after the experimental run, the cover remained at the original position on the manhole top without forming any air leakage from the edge. The photo sequence of mobile cover motions during the multigeysier regime is presented in Fig. 4 using the case of $D_r/D = 0.5$ and $A_v/A_s = 0$ as an example. The mobile cover danced at $t = 12.33$ to 13.2 s [Figs. 4(a–c)], and at this time, the air pocket front just entered the manhole and kept rising. The cover dancing was attributed to the headspace air pressurization created by the continuous rising of the water column in the manhole.

As soon as the air pressure force under the cover overcame the cover weight, the cover bounced up, bringing extra leakage of compressed air in addition to the ventilation through the cover ventilation hole. The leakage resulted in a sudden pressure drop under the cover, so the cover fell back on the manhole top. This cover dancing process occurred repeatedly. During the first geyser, as the free surface of water column rose at an accelerating speed, the significant air pressurization and direct water impact led to cover blowoff at $t = 13.37$ s [Fig. 4(d)]. Then, the cover fell back onto the manhole top but only provided partial coverage at $t = 13.95$ s [Fig. 4(e)]. The cover blew off again by the second geyser of air–water mixture eruption at $t = 14.62$ s [Fig. 4(f)] and fell onto the lab ground. The cover blow height caused by the second geyser was two to three times larger than that around 0.05 m by the first geyser. The previously mentioned cover motions agree with that reported on the news (Holmes 2017; Fox News 2018), proving the rationality of the experimental results.

Pressure Characteristics under Manhole Cover

Fig. 5 presents the evolution of the pressure head h_c beneath the fixed manhole covers. Some pressure oscillations were observed at the beginning of each case, and they should be induced by the initial pressure difference across the valve according to Wright (2013)

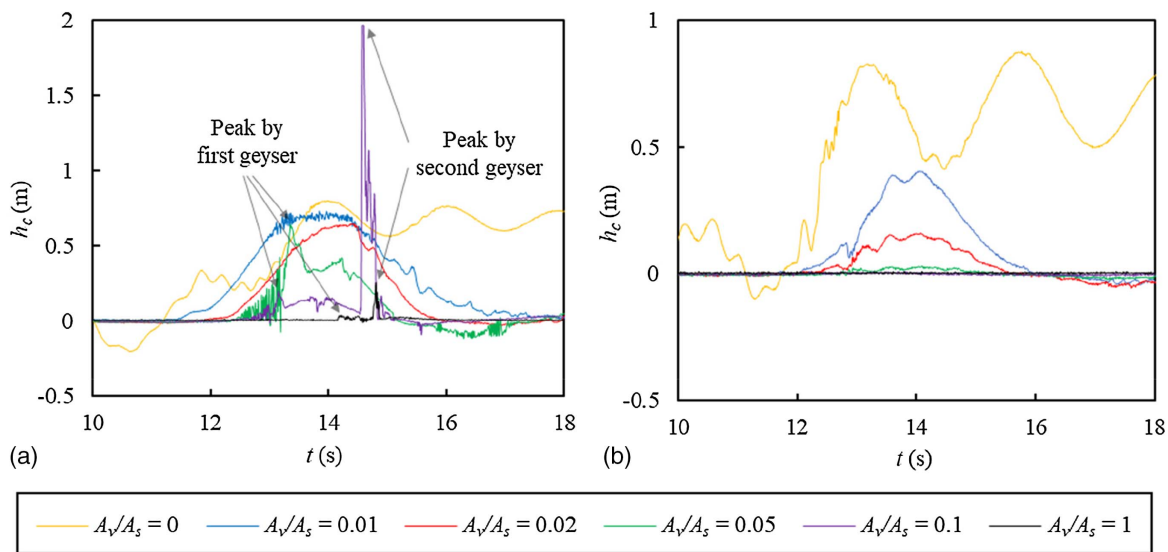


Fig. 5. Evolution of pressure head h_c beneath fixed covers for (a) $D_r/D = 0.5$; and (b) $D_r/D = 1$; the air pockets enter the manholes around $t = 12$ s.

and Chan et al. (2018). In our study, the pressure oscillations always damped quickly before the air pocket reached the manhole, except for the cases with a nonventilation cover. Therefore, the pressure oscillations can be considered to have limited effects on the following air pocket rising and potential cover motions.

After the air pocket rising into the manhole, there are different evolution patterns of h_c for the nongeyser, single-geyser, and multi-geyser regimes.

During the nongeyser regime for the cases of $D_r/D = 0.5$ with minor cover ventilation ($A_v/A_s = 0$, and 0.01) and $D_r/D = 1$ with all cover ventilation conditions, h_c rapidly increased and reached a peak when the air pocket broke through the free surface of the water column above. Then, h_c oscillated with decaying peak magnitudes. Finally, h_c tended to 0 after a long-period oscillation stage, except for $A_v/A_s = 0$, with the final pressure reaching the difference between the system external head and the hydrostatic head of the water column remaining in the manhole because of the zero ventilation of the cover. These pressure head evolutions verify the numerical results of Wang and Vasconcelos (2018, 2020).

During the single-geyser regime for the cases of $D_r/D = 0.5$ with $A_v/A_s = 0.05$, the largest peak of h_c appeared significantly sharper than that for the nongeyser regime, and it was obtained as soon as the geyser of water column eruption caused by the air pocket rising in the manhole. Following the largest peak, h_c oscillated and finally tended to 0. For the cases of $D_r/D = 0.5$ with $A_v/A_s = 0.02$, the pressure evolution was similar to that of the nongeyser regime. This is because the air pocket broke in the manhole, and after that the water kept rising and impacted the cover due to inertia, generating a weak geyser.

During the multi-geyser regime for the cases of $D_r/D = 0.5$ with large ventilation ($A_v/A_s = 0.1$, and 1), the peak of h_c as soon as the first geyser of water column eruption happened was followed by another peak that was eight to ten times higher and caused by the second geyser of air–water mixture eruption.

Fig. 6 shows the evolution of h_c beneath the mobile manhole covers. Compared with the fixed covers, the most remarkable differences of the mobile covers were pressure oscillation with high frequency, significant decline of pressure peak magnitude, and pressure drop after geysers that even reached apparently negative

pressures. The pressure oscillations occurred after the air pockets rose into the manholes and ended at the air pockets breaking (nongeyser regime) or water column ejection and cover blowoff (multi-geyser regime). The pressures oscillated with frequencies of 10–20 Hz, which were potentially influenced by manhole diameter, cover ventilation, system external head, and initial air pocket volume. Cover dancing is supposed to be the trigger because the dancing frequencies match the frequencies of pressure oscillations. Using a mathematical model, Tijsseling et al. (2019) obtained a manhole cover dancing with a frequency of 12 Hz, and the magnitude agrees with our result. The pressure peak declination was caused by cover dancing and blowoff and therefore the loss of their ventilation restriction for manholes. The sudden pressure drop is believed to have been triggered by the fluid's (air, water, and air–water mixture) violent ejection out of the top of the manholes. Depressurization was also observed during the geyser events in a prototype stormwater system by Wright et al. (2011a), during the release of air pockets in simplified stormwater systems by Leon et al. (2019) and Wang and Vasconcelos (2020), and during rapid filling of the model of a small portion of a prototype system by Liu et al. (2020). Depressurization possibly causes negative pressures in manholes, and the most significant negative pressures were observed in the cases of $D_r/D = 1$ with mobile covers [Fig. 6(b)]. In these cases, during the water column and air–water mixture rising in the manhole, the cover was dancing; therefore, more air is ejected out of the manhole, and the negative pressure can be enhanced compared with the fixed-cover cases. After the air pocket broke in the manhole, the cover remained at the original position on the manhole top without forming air leakage from the edge. In comparison, in the cases of $D_r/D = 0.5$, the mobile cover blew off due to the water column impact. Because of losing the ventilation restriction of the cover, the air freely entered from the manhole top and weakened the depressurization in the manhole.

The maximum negative pressure heads $h_{cn,max}$ normalized by the manhole diameter D_r are shown in Fig. 7(a). When $D_r/D = 1$, $h_{cn,max}/D_r$ decreased with increasing A_v/A_s for both fixed and mobile covers. When $D_r/D = 0.5$, $h_{cn,max}/D_r$ for mobile covers had limited variation, and $h_{cn,max}/D_r$ for fixed covers first increased and then decreased, reaching a peak when $A_v/A_s = 0.05$.

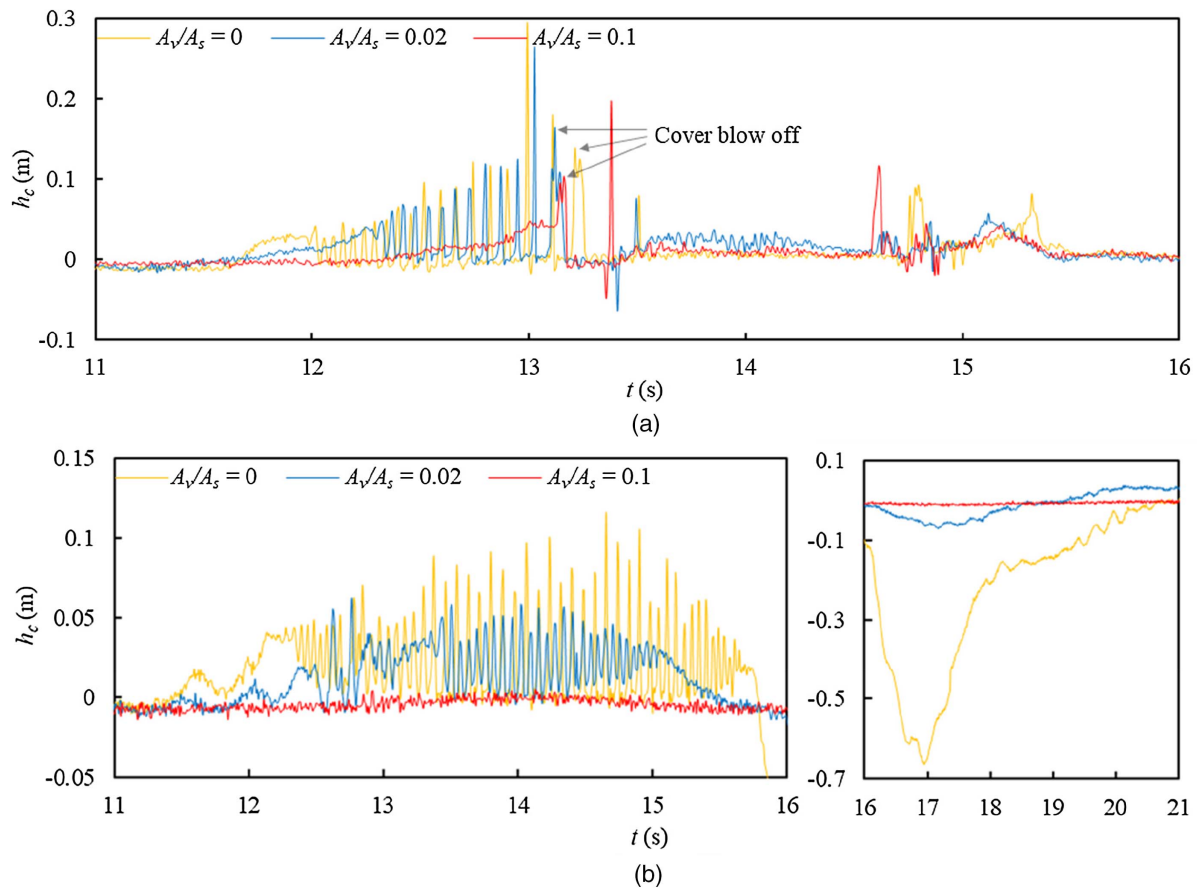


Fig. 6. Evolution of pressure head h_c beneath mobile covers for (a) $D_r/D = 0.5$; and (b) $D_r/D = 1$; the air pockets enter the manholes around $t = 12$ s.

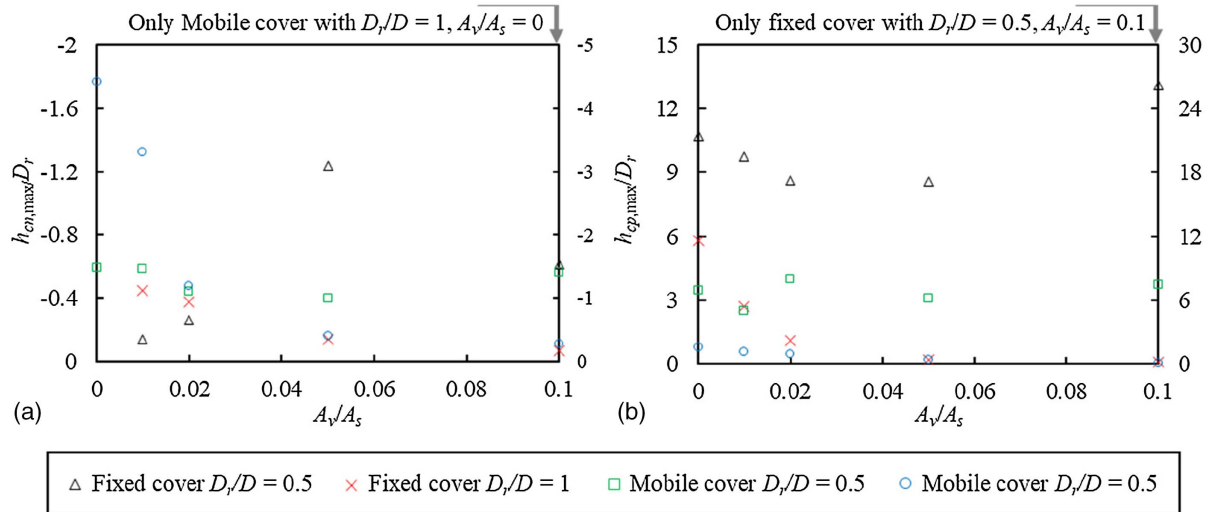


Fig. 7. (a) Maximum negative pressure head $h_{cn,max}/D_r$; and (b) maximum positive pressure head $h_{cp,max}/D_r$ versus ventilation area A_v/A_s .

Except for the condition of $D_r/D = 0.5$ and $A_v/A_s = 0.05$, $h_{cn,max}/D_r$ for mobile covers was always larger than that for fixed covers.

Fig. 7(b) presents the normalized maximum positive pressure heads $h_{cp,max}/D_r$. For the fixed-cover cases, $D_r/D = 0.5$ had larger $h_{cp,max}/D_r$ than $D_r/D = 1$, which verifies the numerical

results of Wang and Vasconcelos (2020). With A_v/A_s increasing, $h_{cp,max}/D_r$ continued to decrease when $D_r/D = 1$, whereas it first decreased and then increased when $D_r/D = 0.5$, with the maximum reaching 26 at $A_v/A_s = 0.1$ (2.5 times the value at $A_v/A_s = 0$) and the minimum around 8 located between $A_v/A_s = 0.02$ and 0.05. Compared with the fixed-cover cases, the mobile-cover

cases have much smaller $h_{cp,max}/D_r$, and the largest difference reached 22.57 and occurred for the condition of $D_r/D = 0.5$ and $A_v/A_s = 0.1$. The main reason is that the cover motions of dancing and blowoff enhance manhole ventilation. The value of $h_{cp,max}/D_r$ for mobile covers also increased with decreasing D_r/D . With variation of A_v/A_s , $h_{cp,max}/D_r$ values fluctuated around 3.5 and 0.3 for $D_r/D = 0.5$ and 1, respectively. The cover motions decreased the ventilation restriction for manholes, therefore bringing about the limited effect of A_v/A_s on $h_{cp,max}/D_r$.

Because the manhole height, system external head, and initial air pocket volume remained constant in our study, a brief discussion is conducted about differences if these parameters are changed. Increasing the height of manholes can cause the rising water column and following rising air–water mixture to hardly impact the covers. Increasing the system external head should enhance the geyser strength and air pressurization underneath the covers (Leon et al. 2019). The experimental results of Wright (2013) indicated that if $V_a/D^3 > \sim 10$, the surge heights of geysers are approximately independent of air pocket volume V_a . In the present study, V_a/D^3 equaled 9.5, which means that the effect of further increasing V_a should be relatively small. Decreasing V_a can reduce the surge heights of geysers, making the covers suffer smaller air pressurization and less significant impact of water and air–water mixture (Wang and Vasconcelos 2020).

Conclusion

In the present study, geysers through covered manholes induced by air pocket release in stormwater systems were experimentally explored. The key findings, including geyser regimes, mobile cover motions, and pressure heads at manhole headspace, are summarized as follows.

Depending on the number of geysers during air pocket release, three regimes are observed: nongeyser, single-geyser, and multi-geyser. The only geyser of the single-geyser regime and the first geyser of the multi-geyser regime are the eruption of a water column driven by air pocket rising in the manhole. The second geyser of the multi-geyser regime is an air–water mixture eruption caused by the large pressure gradient between the manhole and main tunnel after the first geyser. Each geyser corresponds to a sharp peak of pressure head at the manhole headspace.

For fixed manhole cover conditions, both increasing the manhole diameter and decreasing the cover ventilation decrease the geyser number and even prevent geyser occurrence. Compared with the first geyser, the second geyser during the multi-geyser regime has a larger eruption height and volume as well as more violent pressure acting on the cover; therefore, it should be given special attention in practical engineering.

For mobile cover conditions, increasing the manhole diameter can prevent geyser occurrence, whereas changing the cover ventilation has a limited effect in this aspect. Mobile covers can dance and blow off during air pocket release, bringing about the pressure characteristics of oscillation with high frequency, decline of peak magnitude, and significant drop and possibly reaching negative pressures after geysers. The most significant negative pressures are observed in the cases of mobile covers with large manhole diameter and small cover ventilation.

The maximum positive pressure head after air pocket release can be seven times higher for the fixed cover than that for the mobile cover, and the largest difference between the fixed and mobile covers occurs for the conditions of small manhole diameter and limited cover ventilation. This supports the numerical results of Wang and Vasconcelos (2020) that bolting covers to manholes can prevent

cover displacement but might cause structural damage if the structure cannot absorb the effects of air pressurization. The maximum positive pressure head decreases with increasing manhole diameter. Increasing cover ventilation decreases the maximum positive pressure head for fixed covers, whereas it has a limited effect for mobile covers because of the cover motions weakening the ventilation restriction of covers for the manholes.

According to Wright (2021), the results of geyser duration and strength from laboratory experiments are somewhat inconsistent with prototype geysers, and the main reasons are the enhanced influences of surface tension and viscosity as well as constraints of geometry and air pocket volume in small-scale models. For geysers in covered manholes, laboratory experiments can also be affected by scale effects in view of air pressurization underneath the covers and cover motion characteristics. Therefore, experiments performed in larger-scale models and comparisons between laboratory and prototype results are suggested in the future for improving upon the findings in the present study.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

Acknowledgments

The authors acknowledge the financial supports of the National Natural Science Foundation of China (Grant Nos. 51809079 and 51779082), the Natural Science Foundation of Jiangsu Province (Grant No. BK20201315), and the Fundamental Research Funds for the Central Universities (Grant No. 2019B18414).

Notation

The following symbols are used in this paper:

- A_s = cross-sectional area of manhole;
- A_v = ventilation area;
- D = diameter of horizontal tunnel;
- D_r = manhole diameter;
- H_o = system external head;
- h_c = pressure head below manhole cover;
- $h_{cn,max}$ = maximum negative pressure head below manhole cover;
- $h_{cp,max}$ = maximum positive pressure head below manhole cover;
- t = time after opening valve; and
- V_a = initial air pocket volume.

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